

(git/github)

↳ Github for Windows (installation)

↳ git for all platforms:-

Setup:-

To configuring user information used across all local repositories:-

```
git config --global user.name "[fname LName]"  
git config --global user.email "[valid-email]"  
git config --list (To see what we implement till now) details:-
```

Note:-

*) Git command also be implement on bash terminal. But, if use it with code editor then it will be more beneficent. So, we use git process and coding process together easily.
we use VS code:-

*) Inserting commands through terminal

Basic Git commands

a) Clone and setup

Clone: cloning a repository on our local machine.

Remote

[github] folder,
files.

local

Laptop/Computer's files:

So, clone command is being used to copy our github repo on our system.

Command:- (folder link)
git clone (repo link)
(direct link)
ory

we took repo from git hub by using git clone command. We copy it to our vs code editor by type in terminal:-

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(Fly away from topic)

In github, changes known as
commit:-

In github, when we change (commit)
file will be updated:- everytime,
Like it take && and saved it in
git-repo:-

Note

In readme.md file, we use
special syntax (HTML):-

Like for sent updated message
to next line, make new file for
every new change, we use HTML
commands;

For example

or like it?

~~Hello~~ initial commit.

Hello, My -- is Awaiz

Hello! My name is Awaiz

(br)

if I change it

I'm 15 yrs old.

Hello! myName is Awaiz

it write on other

I'm 15 years old. ---

line:

it saved it like

Hello I My -- is Awaiz. I'm 15 yrs old

(So, comeback now to
again clone topic) :-

After completed our cloning
process, if we want to do
something with directory, folder,
file. we use:

(cd) command

Syntax

cd repo/d/f name

After, this we go inside of our
folder. Like. Apna-Awais as

cd Apna-Awais

① For clean terminal, we use clear command

a. if we want to see all files in our

② folder we use ls command :- list files :-
→ it is for normal files which I can see.

③ For Hidden files (which access by terminal)
we use ls -a commands :-

Status Command

Tell us the status of the code

git status

if data in remote are local is same;
then, us give us

On branch main

Your branch is up to date

code in github	=	code in code editor
-------------------	---	------------------------------

And, if I change something in
local. Then both readme.md files
are different now. So, if we check
status with ^{out} commit my local
file

Then, it will show long message,
that changes not state for
commit

After modifying the local file,
we have to do 2 steps for
add or commit to remote.

add → commit
(1) (2)

- So, Like files, it is not sufficient to code. In local, if I add another file other than (readme.md) without step verification to remote:-

it shows a message there
untracked file error popup

means files is not tracked by
git bcz you create it in
local.

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(When we use git status,
we use 4 types of status)

untracked

(new files that git doesn't yet track)

modified

(changed code
or content of

staged

file)

(file is ready to be committed)

when we add data it becomes

staged)

unmodified

(unchanged)

Add & Commit

add - add new or change files in your working directory to the Git staging area.

git add <-file name ->

for add all at once **git add.**

Commit - it is record of change

git commit -m "some message"

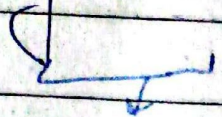
So, after these data will be committed but not show on our Github window. For this we use:-

Push Command

(push → upload local repo content to remote repo)

git push origin name

git push origin main



basic

command



By default gitup repos'
(remote repo's)



out of these repositories we;
here, we talk about that repo;
which is cloned by us in beginning.



So, origin is name of
that github repo copy.



-- (main)

is the branch of
our repo.

NOTED POINT

So, now we see how
we can make project on Github. How
we clone them on code editor. How
we see status, how add, commit our
changes and push them to Github again.

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But, if we start our projects on the local machine, so, how can we manage our local machine folder which we create like in VS code to the Github. For this purpose, we can use:

Init commands

init - used to create a new git repo.

git init

Firstly, we will create a new directory in local:-
we use a command:

(mkdir D.Name)

After create new directory LocalRepo we will go inside it `[cd] localRepo`

a) for going inside of directory, we use `[cd]`

b) for going outside we use `[cd..]`

-) So, now it is a simple file, we should have to make it git file. we can verify that it is or not by `ls` command. that it is git or not.

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if it is not make it by `git init`

Now, we can add files in it like

`index.html`

`color.css`.

↳ So, when we check their status:

it deals them as `untracked files`

↳ So, we add them `git add.`

So, their status will be changed:-

↳ Then commit them `git commit -m "msg"`

↳ Now, we want to upload them to Github.

For this we have to go to Github
create a new repository on
which we have to push our
code

Before push project to Github, we
have to run a command.

`git remote add origin <-link>`

we want to add
new remote git
repo

By default

for Github.

So, our origin is done now

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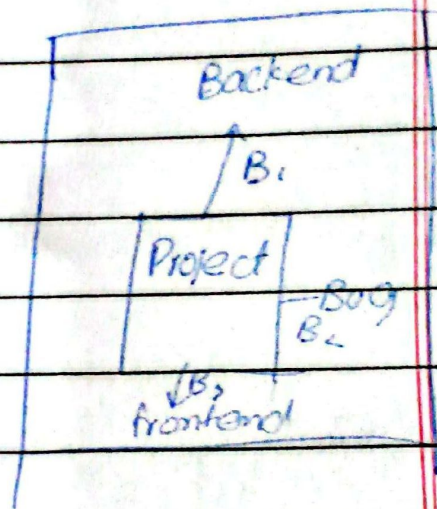
DATE: _____

To check which origin is set we use:

`git remote -v` (to verify remote)

To check on which branch we are:

`git branch`



To rename branch

`git branch -m main`

`Push code`

`git push origin main`

if we have to ~~re~~ work on main branch for whole of project and we don't want to highlight again again origin main, origin main. kahi dard kahi hai, in

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`git push -u origin main`

↓
for set
upstream

We tell it only one time and
after this time we will write only

`git push`

↳ So, files are successfully pushed:-

SHENI

if i want to add new file

README.md-

So, it's a new change (verified by see status;
on tracked)

↓

`git add` (add successfully)

↓

`git commit -m "Add README"`

↓

`git push`

only:- due to -u

Work flow:- Local Git

github repo



clone



Changes
(add + commit)



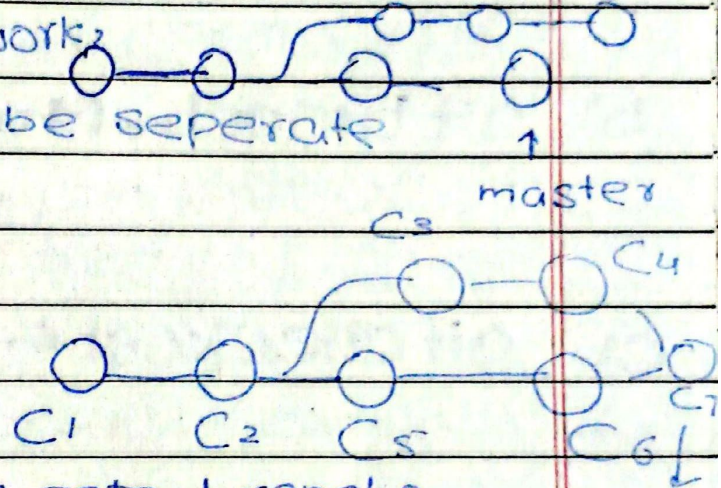
push to github

Git

Branches

↳ if on a product, several people work on it or on several feature multiple features. So, for each

separate work, there must be separate branch.



• Why we create branch?

if one developer work on a specific feature and other

one work on another feature, So question is that the original developer should wait for other to complete their works then he will add new details. So, every developer make their own copy (branch) and then merge their code with main branch.

one work on another

feature, So question is that the original developer should wait for other to complete their works then he will add new details. So, every developer make their own copy (branch) and then merge their code with main branch.

Branch Commands

a) `git branch`

To check branch.

b) `git branch -M main`

To rename branch

c) `git checkout <-branch->`

(To navigate) for going one to
another branch ^{name}

d) `git checkout -b <-branch->`

To create a new ^{new} branch ^{name}

e) `git branch -d <-branch->`

for delete branch. ^{name}

NOTE

Let suppose, we have 3 branches, (main, feature1, feature2). if I want to delete feature1. Then, I should be not on feature1. I should be on another branch except feature1 to delete feature1 branch.

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↳ Data which is shown in feature 2 branch cannot show in main branch.
For this, we can use merging command

Merging Command

way 1

git diff ^{branch} <- name -> Create a PR
git merge ^{branch} <- name ->

way 2

↳ (to compare commits, branched, files and more)
↳ (to merge 2 branches).

PR

Pull ⇌ Request

It lets you tell others about changes you have pushed to a branch in repository on github.

So, By PR when it is updated on Github. By default in local machine in branches code is not merged for this we use:

Pull command:-

`git pull origin main`

used to fetch and download contents from remote repo and immediately update the local repo to match that content.

Merge Conflicts

(An event that takes place when git is unable to auto resolve differences in code b/w two commits)

if we change 1st 2 line contents of both branches and try to merge it. VS doesn't resolve it:-

(changing occurs at some place in two files. So git confuse that which git commit was to be finalized.

Undoing changes

Case 1:-

Staged changes:-

```
git reset <-filename->  
git reset
```

committed change (for one commit)

```
git reset Head~1  
↓  
latest commit
```

committed change (for many commit)

```
git reset <-commit hash->  
for change in git reset --hard <-commit hash->  
VS (local also)
```

git log to check commits:-

Fork

* It's a rough copy.

it is a new repo that
shares code and visibility settings
with original "upstream" repo.